

Organic aerogel as electro-catalytic support in low-temperature fuel cell

Siti Hasanah Osman¹ | Siti Kartom Kamarudin^{1,2}  | Sahriah Basri¹  | Nabila A. Karim^{1,2}

¹Fuel Cell Institute, Universiti Kebangsaan Malaysia, Bangi, Selangor, Malaysia

²Department of Chemical and Process Engineering, Faculty of Engineering and Built Environment, Universiti Kebangsaan Malaysia, Bangi, Selangor, Malaysia

Correspondence

S. Kamarudin, Fuel Cell Institute, Universiti Kebangsaan Malaysia, 43600 UKM Bangi, Selangor, Malaysia.
 Email: ctie@ukm.edu.my

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Summary

Fuel cells have been developed efficiently in recent years as a secure and high-energy conversion technology. The innovation in electrocatalysis focuses on initiatives to minimize or replace Pt electrodes used. As an alternative, organic aerogel materials are normally used as electrodes in low temperature fuel cells due to their high electronic conducting and electrocatalytic characteristics. These are usually characterized by fundamental functional elements of strong covalent C-C bonds from organic precursors. This review will mainly focus on organic aerogel nanomaterials related to carbon aerogels, carbon nanotube aerogels, and graphene aerogels as electrocatalytic supports by surface modification or structural features in catalysis. In contrast, organic aerogel materials prove to be durable, economical, and active electrocatalysts that can provide outstanding electrocatalytic efficiency in low temperature fuel cell applications. Last, this paper recaps the challenges faced in current fuel cell applications and the use of organic aerogel materials to address these challenges.

KEYWORDS

carbon, carbon nanotube, electrocatalyst, graphene, low temperature fuel cell, organic aerogel

1 | INTRODUCTION

In the 21st century, fuel cells will receive very wide attention, especially in the energy industry, due to their environmentally friendly and sustainable nature. Almost all types of fuel cells will be introduced commercially in the industry and become an attraction in the field of research. Thus, it currently leads to a new dimension in energy production. Currently, Figure 1 shows the development of hydrogen-based fuel cells as fuel and have been monopolized compared to other fuel cells.¹ Fuel cells can be categorized into several types depending on the type of fuel and electrolyte used. Fuel cells consist of several important parts that function significantly to ensure that they operate optimally.^{2,3} Overall, low

temperature takes place in the range of (<300°C). Hence, this review is focused on organic aerogel in low temperature for fuel cells polymer electrode membrane fuel cells (PEMFCs), direct methanol fuel cell (DMFC), and microbial fuel cell (MFC) on improving electrocatalysts.

Platinum (Pt) is used in fuel cells to branch the chemical reaction of hydrogen with oxygen. In general, Pt is a good inert electrode because it is chemically unreactive and is very beneficial for fuel cell use. Platinum does not participate in redox reactions because it can easily adsorb hydrogen.³ Platinum (Pt) has the highest activity in both anodic and cathodic reactions.⁴ The platinum-based catalyst is the most efficient in the methanol oxidation (MOR) reaction. Based on the high level of platinum consumption used in fuel cells, affects the increasing cost of

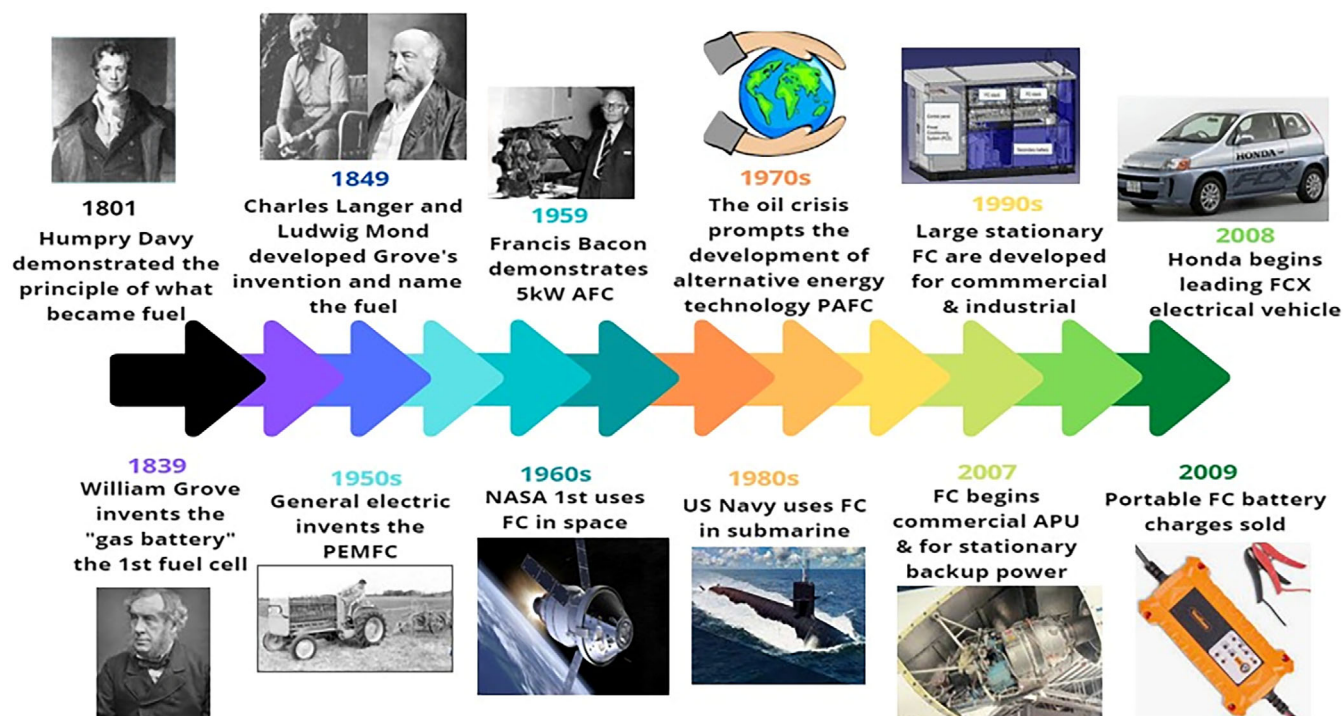


FIGURE 1 History of fuel cell technology

fuel cell systems as well as the performance and stability in terms of poisoning and sintering.⁵ In the research world, bombastic studies relating to economic and high-powered alternatives to catalyst development are gaining momentum. To have a positive impact on electrochemical reactions, the catalyst layer should be fabricated and produced optimally. Karim et al.⁶ stated that to classify the best catalyst, it must be a good compartment for transferring electrons, protons, and reactants to produce a good reaction. This technique can be entirely purified when the structure and composition of the catalyst layer are available under optimal conditions. Regrettably, due to a lack of resources and high costs, the use of this precious metal is a major barrier to the commercialization of PEM fuel cells. To effectively resolve this challenge, researchers must improve the loading and effective use of Pt catalysts on support materials.

Researchers have reacted positively to a novel strategy for organic aerogels that acts as a fuel cell catalyst that was proposed in recent years particularly in electrocatalytic applications, due to their good properties and high performance.^{7,8} Organic aerogels are distinguished by their basic functional units, which are made up of strong C-C covalent bonds derived from organic precursors. As a result, a wide range of exceptional properties, such as controlled chemical synthesis and structure design, as well as porosity of organic material, are possible. Furthermore, when opposed to other catalyst supports, it has

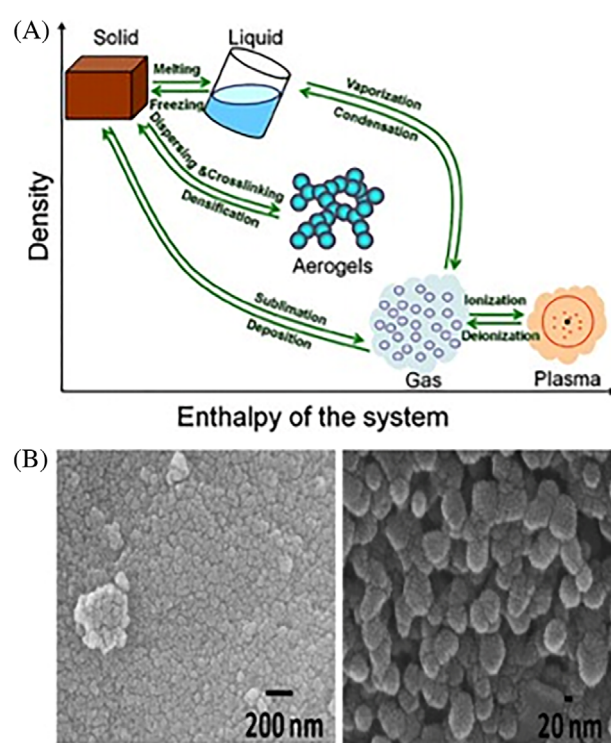
high electronic conducting properties and electrocatalytic properties, making it useful in fuel cell applications.⁹ Organics aerogel are separated into two sections. The first part is carbon-based, including carbon aerogels, carbon nanotube aerogels, and graphene aerogels, while the second part is polymer aerogels, including cellulose and polymer conduction.¹⁰ Gu et al.¹¹ studied the potential in the preparation and application of special features of carbon aerogels that work well in energy applications, particularly in catalytic applications. For the first time in history, utilizing a supercritical drying method, the aerogel former was synthesized by Kistler in 1931. Next, Teichner 1968 and Russo in 1986 developed the second generation of aerogels using user-friendly precursors such as tetramethyl orthosilicate (TMOS) and tetraethyl orthosilicate (TEOS). The results of the study concentrated on inorganic silica aerogels and received less attention to the potential of organic precursors. Pekala et al.¹² suggested a new class of aerogels with the production of resorcinol and formaldehyde precursors, and the aerogel revolution came to the fore and was referred to as organic aerogels. Carbon aerogels can be produced based on aerogel organic pyrolysis heated in an inert atmosphere. The researchers then synthesized carbon nanotube aerogel (CNT) and graphene aerogel from carbon aerogel, which does not require a carbonization process in its manufacture.¹³ Table 1 shows the characteristics of each type of organic aerogel material.

TABLE 1 Classification and their properties organic aerogel

Types	Categories	Properties
Carbon aerogels	Carbon	✓ Textural properties controllable by synthesis conditions ✓ Uniform morphology through the bottom-up process ✓ Applicable to large-scale production
	Carbon nanotube (CNT) Graphene	✓ Crosslinked each other through van der Waals interactions ✓ Promising candidates as electrically conductive materials ✓ The carbonization process is skippable
Polymer aerogel	Cellulose	✓ High mechanical properties can stand with compressive performance without change in shape or fracture ✓ ultimate resource of cellulosic materials come from lignocellulosic biomass
	Conducting polymer	✓ Excellent electrical conducting properties ✓ Reversibility of doping-dedoping endows these conducting polymer potential

Aerogels are an amazing three-dimensional (3D) nanostructure that has a very high pore structure, a high specific surface area, pore size control, an ultralow density, and a very low thermal potential, making this material a good material for thermal insulators and use in various applications.¹⁴ Aerogel nanostructure-related applications have been extensively studied by Lawrence and his team.¹⁵ Aerogels are inhibited by air with a volume of 95% and a very low density compared to other subjects.¹⁶ As shown in the density graph in Figure 2A, compared to the enthalpy of the system, suggesting the aerogel as a new state of matter, an intermediate state between liquid and gas, due to the aerogel density between the conditions of liquid and gas.¹⁷ Aerogels are derived once pores are replaced with air, which comes from the sol-gel, and all the interconnected network structures are maintained without shrinkage. Figure 2B shows the porous and continuous aerogel network structure through electron microscopic images. Different forms of aerogel structures, such as micropores, mesopores, and macropores, can be produced.¹⁸ This is the discovery of a special structure constructed from a network of nanosized particles that are interconnected and provide distinct gains for the fabrication of porous materials.^{19,20}

Interestingly, almost all of the outcomes of research revealed that organic aerogels used as catalyst supports in low-temperature fuel cell applications stabilize nanosized metal particles, resulting in a strong catalytic action.²¹ The texture and pore size resulting in the synthesis method plays a significant role in providing robust catalytic support for fuel cell applications.²² Alegre et al.¹⁸ demonstrated that it is possible to minimize the rate of mass transfer at the catalyst layer based on the special pore texture of the modifiable carbon. The potential uses of organic aerogels have led to studies linked to their synthesis and use in the commercial and scientific.

**FIGURE 2** (A) Different states of matter in the “density” vs “enthalpy of the system” diagram. (B) Microscopic images of the highly porous, continuous 3-D network structure of the aerogel layer

However, there are only a few reviews and books related to organic aerogels. This desire is to be examined in such a way that the study of organic aerogels can meet the requirements of the latest trend demand.

This review emphasizes recent developments concerning the role of organic aerogels in fuel cell applications discussing in detail the method of organic aerogel synthesis as catalysts and catalyst support. This review paper is based on the nominated papers published in the

period 2016–2022. Carbon aerogels, graphene aerogels, and carbon nanotube (CNT) aerogels in low temperature fuel cells are the focus of this review paper. This paper also highlights the recent challenges and expectations of upcoming organic aerogel incorporation for catalytic fuel cell applications.

2 | PRODUCTION METHOD

The success of the aerogel preparation method may produce good material potential. The formation of primary nanoparticles through hydrolysis processes is followed by condensation, where both processes compete during synthesis. The synthesis ends with the removal of pore fluid and the process of formation and drying without disturbing the structure of the network. Previously reviewed, it was noted that production methods for all types of aerogels are generally the same.²³

2.1 | Sol-gel process

Three-dimensional extrusion, known as gel extrusion, is a porous solid network covered by a liquid. Two parameters play a role in this gelation method, namely, pH alteration and sol temperature. Sol-gel is a synonymous method of aerogel preparation and is known for its wet chemical aerogel approach.²⁴ Several aspects of this method need to be considered in the production of the aerogel structure, such as the use of precursors, the handling of the operation, and the appropriate infrastructure for the application to be developed. Organic carbon aerogels are commonly synthesized using various catalysts related to the polymerization of resorcinol and formaldehyde.²⁵ In addition, the use of carbon aerogel precursors in large-scale production can occur, the synthesis mechanism will control the properties of the texture, and the bottom-up process can standardize aerogel morphology.

Recently, apart from conventional polymers, a simple and efficient preparation method was suggested by several researchers, such as Sahoo et al.,²⁶ Zhou et al.,²⁷ and Tesfaye et al.,²⁸ regarding graphene and CNTs as precursors. In addition, the carbonization process can be ignored by using this precursor and making the synthesis process simpler and more economical. Several methods can be used in the production of graphene aerogel, that is, sol-gel, self-support, and substrate. A fantastic attribute in the production of electrically conductive materials can be detected in CNT aerogels. This is due to the interaction of van der Waals, which makes them directly linked to each other in the production of activities. However, there are restrictions on the direct formation of

CNT aerogels. This is due to the resulting 3D frame being, as expected, imperfect, difficult to predict mechanical operation, and minimal elasticity.²⁶ To overcome this challenge, the use of surfactants and the handling of polymeric materials are significant issues in minimizing CNT surface activity and network resistance in the 3D framework.²⁹

2.2 | Drying

Drying is a process of extraordinary importance for the aerogel production process. Additionally, several procedures used in the drying process to achieve wet gel-like supercritical drying (alcohol, acetone, CO₂) are used in the drying process, ambient pressure drying,³⁰ freeze-drying,³¹ etc.,^{32–34} solvents in hydrogels are only transitioning conditions, and a solid framework can be obtained after completion of this process. In ambient drying, the surface tension of the pore capillary expansion to the pore wall is an important aspect to determine the failure of the structure.³² The aerogel structure is successful if there are no defects or failures in the porous structure in this process. These drying products can generally be classified according to drying methods such as aerogels (supercritical drying), cryogels (frozen drying), and xerogels (oven drying). Monolithic samples can be produced from this process.³⁵

2.2.1 | Supercritical drying

The supercritical drying process gives excellent consistency and makes the structure more porous.³⁶ This shows that this technique is more user-friendly, accurate, and good than other wet gel drying techniques because it removes the liquid present in the pores. However, the performance of the final product will have a negative effect if solvent replacement is not complete during this technique. Apart from that, this technique is also not friendly for large-scale commercial production because it is not economical, takes considerable time, is hard and complicated to operate, and produces a high volume of solvent waste. All this occurs, and supercritical drying requires high temperature and pressure to produce a good structure that is dependent on the pore fluid. In this technique, the type of solvent that dissolves on wet gels must be carefully considered when determining the critical drying conditions.^{37,38} Carbon dioxide solvent was introduced as an alternative solution because the critical point was lower than other solvents (TC = 31°C) and convenient with a critical temperature (TC) solvent value of alcohol solvent was higher in ethanol

(TC = 240°C).^{39,40} The critical drying conditions depend on the type of solvent used in the wet gel.⁴¹

2.2.2 | Freeze-drying

In general, freeze-drying is another technique for maintaining porous networks. This method can make the aerogel structure easier, cover low costs, be technically easy to manage, and be environmentally friendly.^{31,32} The process of freezing the liquid starts with freezing the wet gel and is followed by evaporation at very low pressure. Crucial aspects of controlling porous structures are the concentration of precursors and the freezing rate. The final material for this process is known as cryogel and surface porosity with a maximum output of 80%.⁴² The addition of the aerogel structure is only half of the surface area, and the pore volume is smaller than that of the aerogel. Instead, several aspects can be enhanced in this method for the production of the best aerogels, such as the addition of time during aging, the use of low expansion coefficients for solvents, and high sublimation pressure at the pretreatment stage of drying. In addition, the freezing rate associated with ice crystals (shape and size) also influences the aerogel structure. The formation of large crystals will contribute to larger porous aerogels of a size based on a low freezing rate.⁴³

2.2.3 | Ambient pressure drying

Ambient drying is the best option compared to other techniques that are safer and lower cost in large-scale industrial production because ambient temperature plays a role in this technique.⁴⁴ Unfortunately, this technique becomes slightly challenging due to the nonpolar treatment required by the pore walls in the gel for the condensation process. The compression of capillary pressure is

very good for the completion of the condensation process that takes place on the adjacent surface functions. In general, two aspects can be described to ensure that this ambient drying can be used properly. The first capillary pressure reduction can reduce the tension on the solvent surface, where the solvent exchange factor is important. The next important aspect is to use an exchange of solvent. Increased framework strength is used as an option to allow it to withstand high capillary tension.⁴⁵

2.3 | Carbonization

The process of carbonization of the polymer gel process is essential as an additional process in the entire aerogel synthesis, particularly in the production of carbon aerogel. Hanzawa et al.⁴⁶ presented studies on the physical properties of carbon aerogels carbonized under different conditions. Organic aerogels are exposed to high temperatures (above 600°C), as shown in Figure 3. In general, temperatures above 1473 K for the carbonization process will diminish the specific surface area of aerogels due to the excessive density of skeletal carbon. Furthermore, significantly higher electrical conductivity can be attained by completing graphitization performed at a temperature above 2273 K.^{48,49} Electrochemical applications require high electrical conductivity to survive.⁵⁰ To accomplish these needs, an appropriate temperature is required to obtain a large specific surface area for aerogels. Luckily, aerogel-based graphitics such as graphene or CNTs can be skipped as they are graphitized. Physically or chemically, carbon aerogels with potassium hydroxide (KOH), sodium hydroxide (NaOH), carbon dioxide (CO₂), and water (H₂O) have been activated to produce extra micropores and increase the surface area.^{51,52} Furthermore, several parameters, such as surface area, pores, and product pore size distribution, will be controlled by the activation of carbon aerogel.⁵³

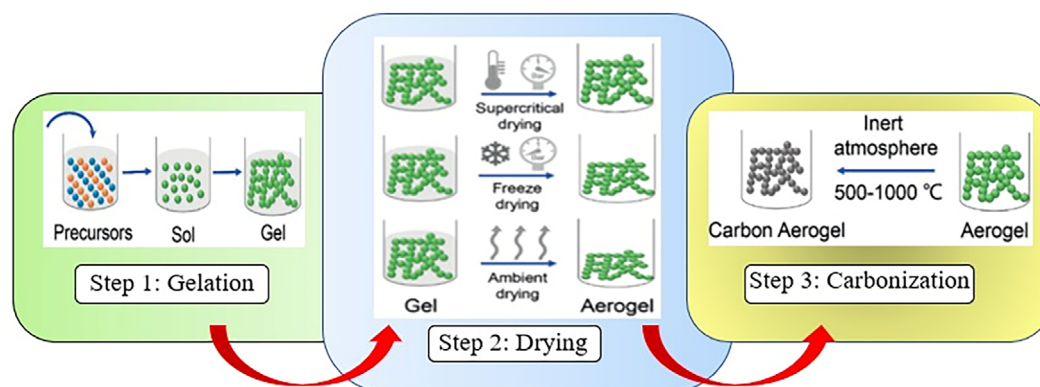


FIGURE 3 Different procedures to prepare aerogel and aerogel composite-based catalysts⁴⁷

3 | ORGANIC AEROGEL APPLICATION IN LOW TEMPERATURE FUEL CELL

Mostly, organic aerogels are suitable for use as electrode materials in fuel cells. Salvador et al.⁵⁴ presented a green approach to producing carbon aerogels where studies on the cathode have been conducted and good conductivity, as well as catalytic results, have been completed for microbial fuel cell applications. It is amazing where less fragile than inorganic aerogel which will squeeze once compressed. In addition, it can stabilize the leaf and is reported as an extremely light material. Based on this feature, it has a great chance to be used in catalytic fuel cell applications.⁵⁵ Table 2 shows some organic aerogel in low temperature fuel cell applications along with their properties.

3.1 | Polymer electrode membrane fuel cells

Generally, carbon which is used as a support for PEMFC catalysts, the size of the pores cannot be changed, but it can be changed in aerogels. Controlled chemical synthesis and design of structure and porosity of organic materials are among the exceptional properties of aerogel materials.⁶² Jiang et al.⁶³ facile reduction method was used to make carbon aerogel supported platinum-nickel solid solution alloy catalysts (Pt₃Ni/CA) with uniform distribution and small particles (3.5 nm). The optimized Pt₃Ni/CA catalysts for ORR have mass activity (MA,

0.25 A mg_{Pt}⁻¹) and specific activity (SA, 4.69 A m⁻²) nearly 3.8 times and 6.5 times that of state-of-the-art commercial carbon-supported platinum catalysts (Pt/C, 0.07 A mg_{Pt}⁻¹ and 0.72 A m⁻²) in rotating disc electrode (RDE). Gu et al.¹¹ introduce the creation and advancement of a porosity-tunable carbon aerogel (CA) as a carbon support option in the PEMFC as a result, in fuel cell testing, it corresponds up to a 713% higher specific power density. Singh et al.⁶¹ recommended carbon aerogel as platinum support by using the non-aqueous medium, resorcinol, and furfuraldehyde were used to polymerize sol-gels, which were then gelated at high pressure. In PEMFCs (H₂-O₂) single cells, a membrane electrode assembly fabricated with the prepared Pt/carbon aerogel catalyst as a cathode and anode is tested, yielding a power density of 536 mW cm⁻² at 0.6 V at 60°C under atmospheric pressure. Furthermore, Öner and team⁶⁰ described PEM fuel cell catalyst support, and 3D graphene aerogel (GA) was synthesized using a self-assembly hydrothermal process. Pt/GA and polydimethylsiloxane, PDMS is the best candidate for being an ORR catalyst in fuel cell applications, according to the results of cyclic voltammetry (CV), electrochemical impedance spectroscopy (EIS), and durability tests illustrated in Figure 4.

3.2 | Direct methanol fuel cells

The mass transportation of fuel and exhaust will be affected by the pore structure in the cell electrode, affecting the overall performance of the cell. DMFC is a type of

TABLE 2 Properties of some organic aerogel in low temperature fuel cell

Aerogel Name	Properties	Application	References
Rh-decorated three-dimensional graphene aerogel (Rh/3D-GA)	Large electrochemically active surface area Long-term durability toward methanol oxidation, high mass activity	Direct methanol fuel cell	[56]
Rh/G-CN aerogel	Large electrochemically active surface area, excellent conductivity	Direct methanol fuel cell	[57]
Pd/GA	High poison tolerance, high stability,	Direct methanol fuel cell	[26]
L -cysteine GA	Large specific surface area, highly efficient charge transfer process	Microbial fuel cell	[58]
GA	High electrocatalytic activity, excellent antipoisoning ability	Direct methanol fuel cell	[59]
Pt/GA and Pt/PPy-GA	More durable toward carbon corrosion, the best electrocatalytic activity for ORR	Polymer electrode membrane fuel cells	[60]
CA	High mesopore volume, higher specific power density	Polymer electrode membrane fuel cells	[11]
CA	High mesopore volume, excellent durability	Polymer electrode membrane fuel cells	[61]

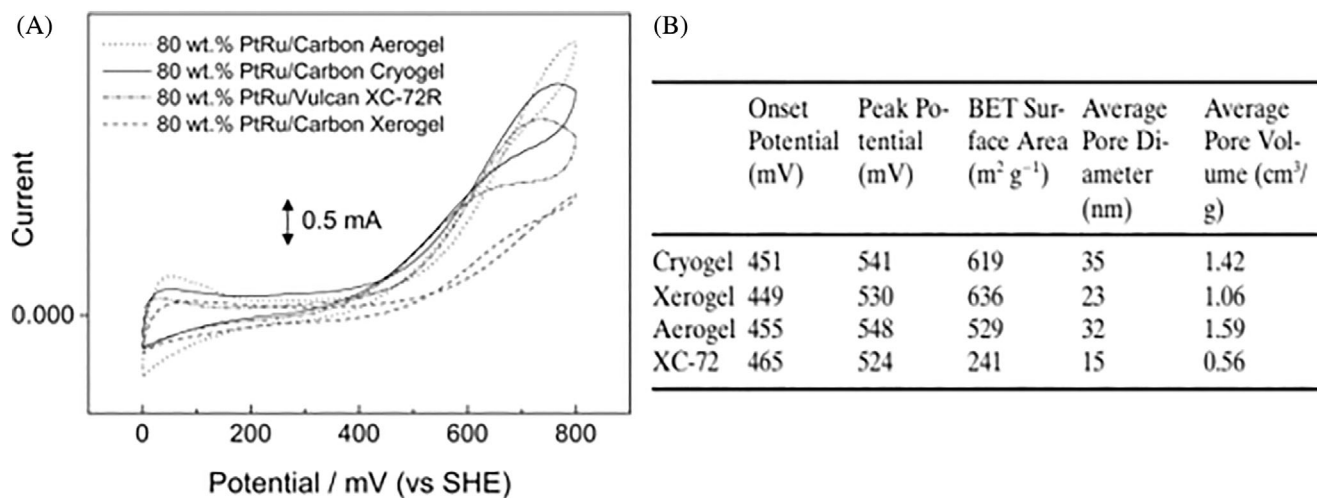


FIGURE 4 Graphene aerogel in ORR performance of polymer electrode membrane fuel cell⁶⁰

low temperature fuel cell that uses methanol as the primary fuel and emits carbon dioxide as an exhaust.⁶⁴ Overall, alternative support materials in DMFC are desperately needed to solve complex two-phase flows. Bong and his team successfully produced carbon aerogels from 2019 to 2020 for direct methanol fuel cells. Bong et al.⁶⁵ formulated carbon aerogel loading with platinum in an anode electrode by a polycondensation reaction of a resorcinol-formaldehyde (RF) polymer that maintains the framework of the pore skeleton with supercritical drying. Bong et al.⁶⁶ incorporated three simple combination methods to develop nanocarbon-based supercritical, freezing, and thermal drying process methods for resorcinol-formaldehyde with their strongly charged DMFC PtRu anode electrocatalysts. The cyclic voltammograms in Figure 5A and several distinguishing parameters in Figure 5B show that the PtRu/carbon aerogel was a large exceptionally charged PtRu nanoparticle and small metallic Pt agglomerate.

The investigation by Kolla and his team⁶⁷ customized the sol-gel system of carbon aerogel and carbon-supported industrial platinum ruthenium catalyst in which fabricated catalyst framework nanoparticles made of metal oxide were heat-treated at 350°C and 600°C. The customized methanol oxidation rate for CeO₂-doped PtRu/C was three times greater than that for unaltered PtRu/C. Fort et al.⁶⁸ produced high mesoporous CAs manufactured utilizing a sol-gel process followed by supercritical drying in an inert atmosphere with liquid CO₂ and thermal pyrolysis. CA has a surface area of 843 $\text{m}^2 \text{g}^{-1}$, and 6% w/w PtRu nanoparticles in geminated exhibit good CH₃OH electrooxidation efficiency. According to Du and his colleague,⁶⁹ cell output was superior owing to smooth mass transport based on a carbon aerogel mesoporous structure as a PtRu catalyst that

demonstrated very small pores with diffusion centered at approximately 11.4 nm. Further investigation by Zhu and his team⁷⁰ studied microwave-assisted polyol by recorded operation with formulated Pt-Au nanoparticles sponsored on carbon aerogel at a 2:1 ratio. The findings show that Au-modified Pt has better electrochemical catalytic behavior and methanol oxidation stability than the traditional Pt/C catalyst of Pt charging. Guilminot et al.⁷¹ studied Pt/CA carbon aerogels synthesizing cellulose acetate-based gels with pyrolysis for use in PEM applications. The results of the study found that Pt/CA based on cellulose acetate can be used as an alternative, and the achievement of ORR is comparable to Pt/Vulcan XC72R. In addition, Guo and coworkers⁷² discovered the GR-CNT aerogel, which has been established as a helpful matrix for ultrafine nanoparticle design. Chronoamperometry tests the Pt/CA catalyst electrochemical activity for methanol oxidation and stability by presenting Pt/CA catalyst density vs time curves and Pt/C catalyst with a steady potential of 0.65 V per 1000s. Yuan et al.⁷³ reported the building phase on the carbon aerogel microstructure, and the catalytic properties of a CA-supported Pt catalyst for methanol electrooxidation were analyzed. The findings show that the drying of ambient pressure contributes to CA with micropores, whereas the CA formulated by supercritical CO₂ extraction (SCE) is predominantly macroporous and has a greater specific surface area, pore-volume, and average pores.

Graphene is a 2D nanostructure with an enormous surface, electronic conductivity, mechanical stability, and low density.⁷⁴ 3D nanostructured aerogels of graphene have recently been formulated by self-assembly of 2D graphene nanosheets.^{75,76} Generally, as shown in Figure 6, the facile method preparation of graphene aerogel was established with a suitable optimum mixture of

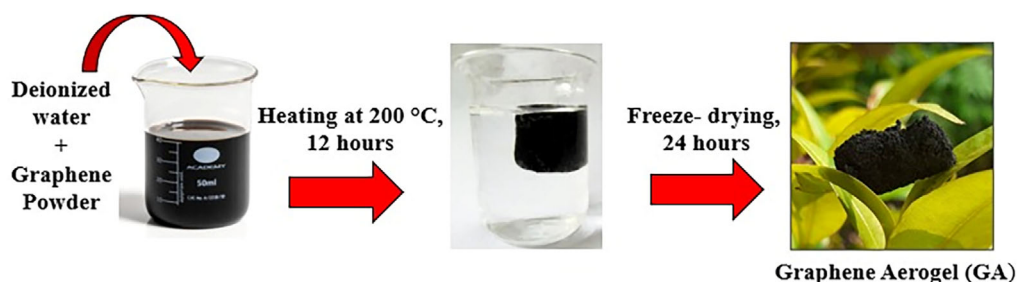
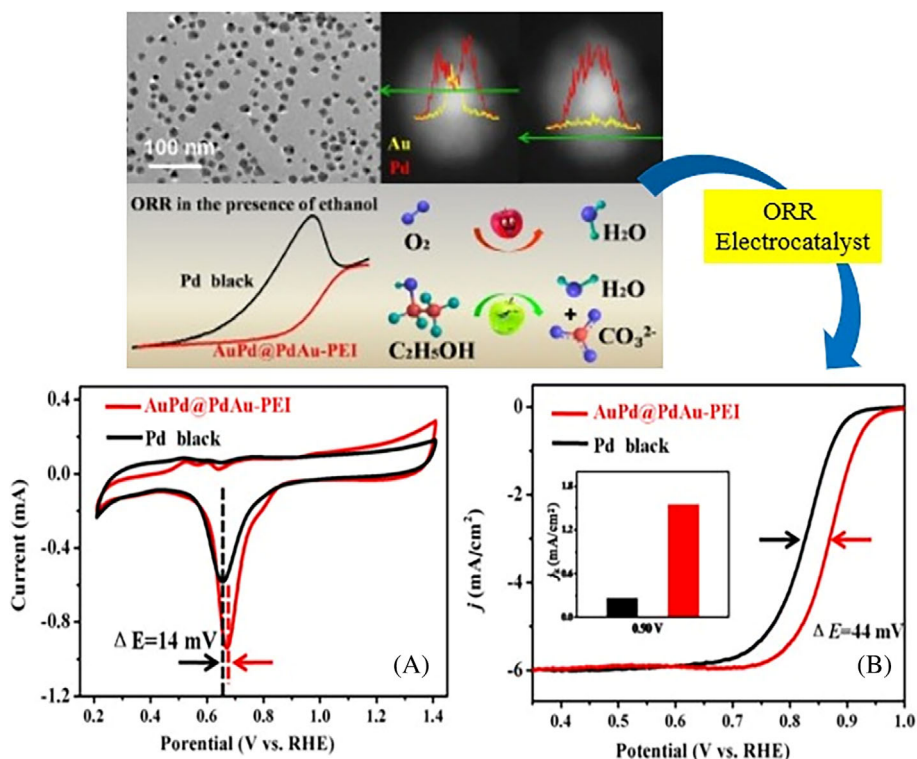


FIGURE 5 (A) Cyclic voltammograms of these catalysts: 1 M H_2SO_4 + 1 M CH_3OH , 25°C, scan rate: 20 mVs^{-1} (B) details regarding starting potential, peak potential, active electrochemical surface area, mean pore size, and the average pore volume⁶⁶

FIGURE 6 Facile preparation hydrothermal process for graphene aerogels



graphene powder and deionized water using a hydrothermal process by modifications relying on Lim and team.⁷⁷ Surprisingly, graphene aerogel allows standing on tree leaves. This demonstrated the extraordinary lightness characteristics of graphene aerogel.

Based on the outstanding properties of graphene aerogel, Yuan et al.⁷⁸ used graphene aerogel essential hydrophobicity and microstructure to create an innovative two-phase anode mass transfer system. The findings show that the anode gas diffusion layer based on graphene aerogel is better than carbon black and poly(tetrafluoroethylene) because of the increased microstructure, which leads to good development in gas-liquid mass transfer management. Furthermore, there was a more than 20% increase in terms of output power density as well as an increase in operating methanol concentration without an additional structure.

Interestingly, Cogenli et al.⁷⁹ studied the diffusion of Pt nanoparticles, nitrogen (N), and boron (B) heteroatom-doped graphene aerogel support products to improve electrocatalytic activity via a microwave heating cycle. The outcomes of electrochemical chronoamperometry analyses indicate that Pt/BGA displayed perfect stability for methanol oxidation and demonstrated outstanding electrocatalytic behavior in cyclic voltammetry against methanol oxidation. According to He et al.⁵⁹ studied a hybrid platinum-graphene aerogel using three combined processes: electrospraying, freeze casting, and solvothermal. Therefore, a mass activity of 1098.9 mA mg^{-1} was recorded for methanol oxidation and is a well-known antipoisoning ability, where the dispersion of Pt particles on the microspheres of graphene aerogel is better than that of graphene oxide and supports commercial carbon black.

Yang et al.⁵⁶ represent a simple and economic bottom-up fabrication of Rh-decorated three-dimensional graphene aerogel (Rh/3D-GA) networks as efficient DMFC anode catalysts using a self-assembly method. The Rh/3D-GA hybrid architecture that results has superior electrocatalytic properties for the methanol oxidation reaction, including a large electrochemically active surface area, low onset potential, high mass activity, and outstanding long-term stability. Other research studies by Sui et al.⁸⁰ studied the zeolitic imidazolate framework ZIF-8 and developed in situ hydrothermal reconstructions that were successful in rendering N-doped graphene aerogels. The high catalytic performance of the Pt-based catalyst for methanol oxidation reaction (MOR) operation is essentially 1.8 times higher than industrial Pt/C, and its long cycle reliability is improved by approximately 18.7% compared to traditional Pt/C. Zhao et al.⁸¹ appropriately manufactured a 3D porous graphene aerogel (3D-NGA) utilizing mixed hydrothermal self-assembly, thermal treatment, and the removal of templates. Pt/3D-NGA has a high electrochemical active surface area (ECSA) of $90.7 \text{ m}^2 \text{ g}^{-1}$ and superior catalytic activity and stability compared to Pt/G and Pt/3DGA catalysts. Duan et al.⁸² focused on graphene oxide aerogel (GOA) Pt nanoparticles for methanol electrooxidation that were ready to act as the catalyst support. The findings indicate that Pt/GOA seems to have a microporous 3D structure that not only speeds up mass transfer but also has a broader efficient methanol oxidation region.

Carbon nanotubes (CNTs) have been introduced into the graphene framework, which has reduced defects for electron transfer while stabilizing the composition of the graphene layer,⁸³ resulting in a dramatic increase in electrochemical performance. According to Zhou et al.²⁷ used the freeze-drying method was used to create three-dimensional graphene/carbon nanotube (GR-CNTs) aerogels, with polyvinyl alcohol (PVA) serving as an organic binder. Due to its unique three-dimensional interconnected porous structure and high specific surface area, the Pt/GR-CNTs catalyst achieved exceptional electrocatalytic properties such as high activity, good antipointing capacity, and amazing durability. Further study by Yan et al.⁸⁴ 3D low-defect carbon nanotube/nitrogen-doped graphene hybrid aerogels (Pt/LDCNT-NG) for efficient electrocatalysts for the methanol oxidation reaction can be easily produced using a bottom-up approach. The finding in comparison to commercial Pt/carbon black, Pt/acid-treated CNT, Pt/graphene, and Pt/nitrogen-doped graphene catalysts, the electrocatalyst produced can have a positive effect on enhanced Pt stability, optimized electronic structure, and provide a drastic performance improvement.

3.3 | Microbial fuel cell

Microbial fuel cells (MFCs) could perhaps convert the chemical energy of biodegradable organic matter directly into electricity, making them useful in a variety of fields such as waste disposal, biosensor manufacturing, and renewable energy generation.⁸⁵ Lóránt and his colleague⁸⁶ studied H-type MFCs, and the efficiency of noble metal-free carbon aerogel-based cathode catalysts was investigated and compared to plain glassy carbon cloth (CC) and platinum-containing carbon powder catalyst (PtC). The finding, surface extension by carbon aerogel (CA) increased maximum power density by 34% over CC, to 14.1 W m^{-3} . The performance of nitrogen-doped carbon aerogel (NCA) was increased to 15.7 W m^{-3} . Wang et al. explore⁸⁷ the three-dimensional N-doped porous graphene aerogel (NGA) polymerized on steel mesh (SM) to construct a simple air cathode structure (NGA-x/SM) for use in MFC applications were created using a combination of hydrothermal method and freeze-drying treatment. As a result, Pt-free air cathodes with high-efficiency NGA can be used in MFCs with a maximum power density of 1593 mW m^{-2} . Zhao et al.⁸⁸ applied hydrothermal treatment, dialysis, and freeze-drying were used to make a three-dimensional (3D) hydroxyapatite/graphene aerogel (HA/GA). The HA/GA anode had excellent biocompatibility and graphene conductivity. As an outcome, an MFC with a HA/GA anode produced a maximum power density of 2.38 W m^{-2} , which was 1.83 times that of a GA anode.

High-performance achievements by Kwok and his colleague⁸⁹ suggested Ru@Pt nanoparticles as a porous anode on graphene aerogel. In summary, the composite improved the response rate by raising the anode conductivity to 30.2 mW cm^{-2} , or 18.7 mW mg^{-1} Ru@Pt could be achieved with 1 M MeOH in 3 M KOH. Further investigation by Kwok et al.⁹⁰ introduced the aerogel concept with ultrafine Pt nanoparticles as a porous electrode for MFCs. As described above, the Pt/GA electrode achieved a 358% increase in the specific power compared to Pt/C, where the porous electrode was more conductive to the maximum specific power obtained by enhanced graphene aerogel, making it possible to move through MFC growth. Tian et al.⁹¹ describe a one-pot template-free synthesis of N-doped carbon aerogel (CA) was used, and it was chemically activated with potassium hydroxide (KOH). CA-10KOH's result (mass ratio of KOH to CA was 10) exhibited exceptional performance, with the highest maximum power density ($967 \pm 34 \text{ mW m}^{-2}$), which was 37.4%, 55.2%, and 283.7% higher than platinum (Pt/C) respectively. The general discussion on organic aerogel shows that the manufacturing methods

used in aerogel preparation and precursor materials are factors that influence pores, pore distribution, and networking.

4 | CURRENT LIMITATIONS TO ORGANIC AEROGEL

The development of organic aerogels in terms of manufacture is challenged with some of the difficulties that must be addressed. In that series, this paper deliberates on some substantial limitations to be controlled by particle size and distribution during the fabrication process especially. However, this can be resolved using the methods discussed above.

4.1 | Control of particle size

In general, electrocatalytic studies have only looked at organic aerogel supports. When compared to catalysts supported by commercial carbon, the results of the inspection revealed that those supported by organic aerogels produced better results. In terms of controlling the size of Pt, particles are important to examine, particularly during the synthesis process, because the support in the catalyst greatly influences the use and activity of the catalyst. As a result, the synthesis of Pt-supported nanoparticles is critical for the reaction in the catalyst material depending on the performance in the catalytic process. To characterize the structure of Pt nanoparticles, field release scanning electron microscopy, transmission electron microscopy, and X-ray diffraction can be used.⁹² Barim et al.⁹³ suggested manipulated variables in the advanced decomposition process by in situ supercritical depositions (SCD) technique, associated with thermal annealing and electrochemical dealloying in supercritical CO₂, such as increasing the temperature where the particle size of Pt increased regardless of the CA used. This indicated that the synthesis method and the parameters played are very important in the production of catalysts, and the average metal particle measurement control method can be applied to expand the supporting materials, which have various properties and characteristics. Douk et al.⁹⁴ established an aerogel with a pore size of 11.4 nm by using a self-assembly process, an efficient method for creating a porous three-dimensional network of Pd aerogel. The variation of the 3D aerogel and Pd combination will raise the rate of mass transfer and electrochemical reaction; in this case, it may improve effective mass transfer, active site exposure, and electronic conduction in electrocatalytic activities. According to Wei and his team⁹⁵ they previously manipulated the

carbon aerogel (CA) to synthesize a new type of supported catalyst carbon-based support synthesis by ethylene glycol liquid-phase reduction method. To achieve a uniform dispersion of platinum, carbon aerogels with different pore structures were studied. Furthermore, Pt/CA demonstrated the highest mass-specific current value for MOR when compared to E-TEK 20% Pt/C catalyst (118.4 mA mg⁻¹), with a value of 395.3 mA mg⁻¹.

4.2 | Structural features

The surface functionalization method must be detailed to improve the performance of organic aerogels. In general, the properties of aerogels with very low permeability can be studied in terms of pore size suitable for water vapor/gaseous but not for water molecules.⁴⁵ Carbon aerogels are a type of aerogel that comes in monoliths, beads, powders, and thin films. As a result, this material can be used in adsorption processes and, more specifically, catalysts.^{46,48} The chemical reaction can be triggered by nanopore structures, particularly in organic polymer aerogels. The structural properties of these nanopores provide an opportunity for increased potential in the use of carbon nanotube aerogels for fuel cells when compared to carbon aerogels for fuel cell applications.⁴⁹ The aspect of porosity can be used to evaluate the effectiveness of factors such as molecular accessibility and rapid mass transport via diffusion. Based on this factor, the aerogel has been used as a catalyst material for over 50 y, which results in a response that allows the electrolyte to penetrate the entire aerogel direction through the high pores as well as the aerogel structures mesoscopic pore diameter.⁵⁰ The pore structure of synthesized GA can be determined using N₂ physisorption. The internal structure of aerogel can be determined using FESEM and TEM. Figure 7 shows the N₂ adsorption/desorption isotherms and SEM images for GA fabricated. The pore diameter is typically around 5.8 nm, indicating a mesoporous structure that is well-interconnected.

4.3 | Distribution of Pt nanoparticles

In the DMFC, a 50 wt% catalyst load is required to compensate for the slow oxidation kinetics. However, high catalyst loading contributed to CO poisoning. Aside from that, the primary issue with fuel cell anode catalysts is that the active sites of platinum, palladium, and other noble metals are vulnerable to CO poisoning. It turns out that organic aerogels can help with CO poisoning. According to the EIS findings, CO poisoning on the Pt active sites has been reduced by observing impedance

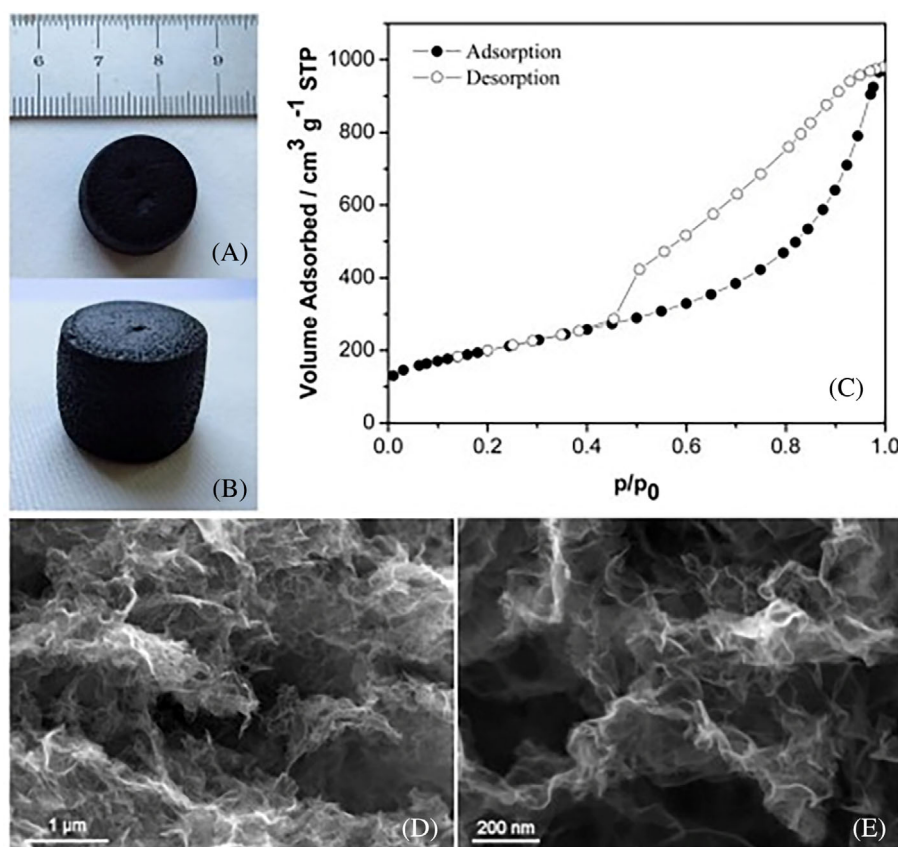


FIGURE 7 Top (A), and side (B) view images of graphene aerogel (GA), (C) SEM isothermal hysteresis loops of GA (D, E) morphology GA⁹⁶

behavior. This case demonstrates how the lower charge transfer resistance (R_{ct}) relates to the higher MOR activity. Such phenomena could be attributed to ultrahigh ESA, improved Pt electronic structure, and excellent mass transfer channels from open structures, which leads to improved Pt electronic structure due to the Pt size effect.⁸⁰ Recent research has sought alternatives and developed a comprehensive Pt/Ru distribution method for the structures involved, particularly the catalyst, in response to this matter. The researcher discovered that the nanostructural support could provide an active reaction in the particle distribution, allowing the catalyst surface to function. He et al.⁵⁹ designed platinum nanoparticles with a narrow size distribution on these agglomeration-free graphene aerogel microspheres to benefit from a microchannel structure having multiple graphene aerogel microspheres. With a platinum engagement of $1098.9 \text{ mA mg}^{-1}$, extremely high mass activity was observed on microspheres supported by graphene aerogels. In terms of antipoisoning ability, this has a positive effect during methanol oxidation. Furthermore, when compared to commercial carbon black supports, the increased number of Pt particles dispersed on graphene oxide provides a significant improvement. Kwok et al.⁷⁷ developed a 3D porous electrode with Pt nanoparticles because the van der Waal strength and hydrogen

bonds are too weak for gelation. The diameter was 2–3 nm with a concentration of 2 mg/mL graphene oxide expressed as MGO₂. Pt particles on carbon black are greater than Pt/C in graphene aerogels, indicating a higher Pt yield. Cögenli et al.⁷⁹ found that the microwave irradiation method can produce platinum nanoparticles decorated at the defect site of the materials, which can provide increased catalytic activity for formic acid electrooxidation in acidic media. Catalytic formic acid electrodes Pt/GA and Pt/GA/C resulted in increased tolerance to CO poisoning effects.

4.4 | Electrocatalytic activity

Based on the fact that methane reforming is a structure-sensitive reaction in which defect sites, edges, and kinks of metal NPs can be active sites, catalysts with smaller particle sizes, which have more metal-support interfacial structures, can have higher activities.⁹⁷ Composite aerogels can be developed by transition metal (Fe, Co, Ni)-doped carbon aerogels, which show high electrocatalytic activity toward oxygen reduction reactions and are suggested as the best candidate for fuel cell applications.⁹⁸ Tesfaye and team²⁸ discovered the maximum power density was 17.5 mW cm^{-2} , and an open-circuit voltage of 0.9 V from

the interaction of Ni and Co bimetal-doped CNT aerogel as an anode electrocatalyst using a polyol reduction and sol-gel method. The characteristics of the metal itself increased the oxidation peak current and decreased the overpotential of the urea oxidation reaction. In addition, the properties of CNT aerogels provide a high surface area and better mass transport due to a 3D network of aerogel structures boosting the electrocatalytic activity. Other researchers are interested in this topic, where Xue et al.⁹⁹ reported the performance of N-GA resulting from pyrolysis at a variety of temperatures, indicating and focusing on cyclic voltammetry. Based on this reported finding, the lowest cathodic reduction peak, featureless slope, and electron transfer numbers higher than other samples, including commercial Pt/C under the same conditions, can be achieved by using N-G-750.¹⁰⁰ Furthermore, a significant test of the catalyst, including the selectivity (eg, CO tolerance and methanol tolerance) and durability, also needs to be considered to measure their performance.^{101,102}

5 | OPPORTUNITIES AND CHALLENGES

Organic aerogels have recently attracted the attention of researchers, and even the industry has noted that many applications have used organic aerogels as one of the components in applications of fuel catalysis. Simultaneous development of material and production process of the material aspects and corresponding production

process to decrease aerogel prices and enable practical applications apart from pure academic research. At the same time, the development of new aerogel types should be application-driven, and target product properties and restrictions should be known from the beginning shown in Figure 8. This can be realized only by dose cooperation between academic groups and industrial partners, which should be encouraged further. The same opportunity can be emphasized to grow organic aerogels.

In terms of opportunities for organic aerogels, polymer- and biopolymer-based aerogels have more recently been explored and emphasized in the research area, especially in catalytic fuel cell applications. Biopolymers are used as precursors from various sources, such as polysaccharides (alginate, cellulose, pectin, chitosan, chitin), lignin, proteins, and others. The advancement of polymer aerogels with exceptional mechanical and thermal insulation via supercritical or ambient pressure drying techniques makes polymer modifications studied, such as polyimide, polyurethane, polyuric, and much more appropriate for aerospace applications in specific applications exhibiting diverse structures and properties.^{103,104} Nanofiller materials such as SiO₂, clay, cellulose nanofibrils (CNFs), carbon nanotubes (CNTs), and graphene or graphene oxide (GO) are often added to the polymer matrix for the production of nanofiller-polymer hybrid aerogels to obtain a strong, cohesive structure in polymer nanopores. The unique feature of aerogels regarding properties as well as their porous network at the molecular level is influenced by biopolymer

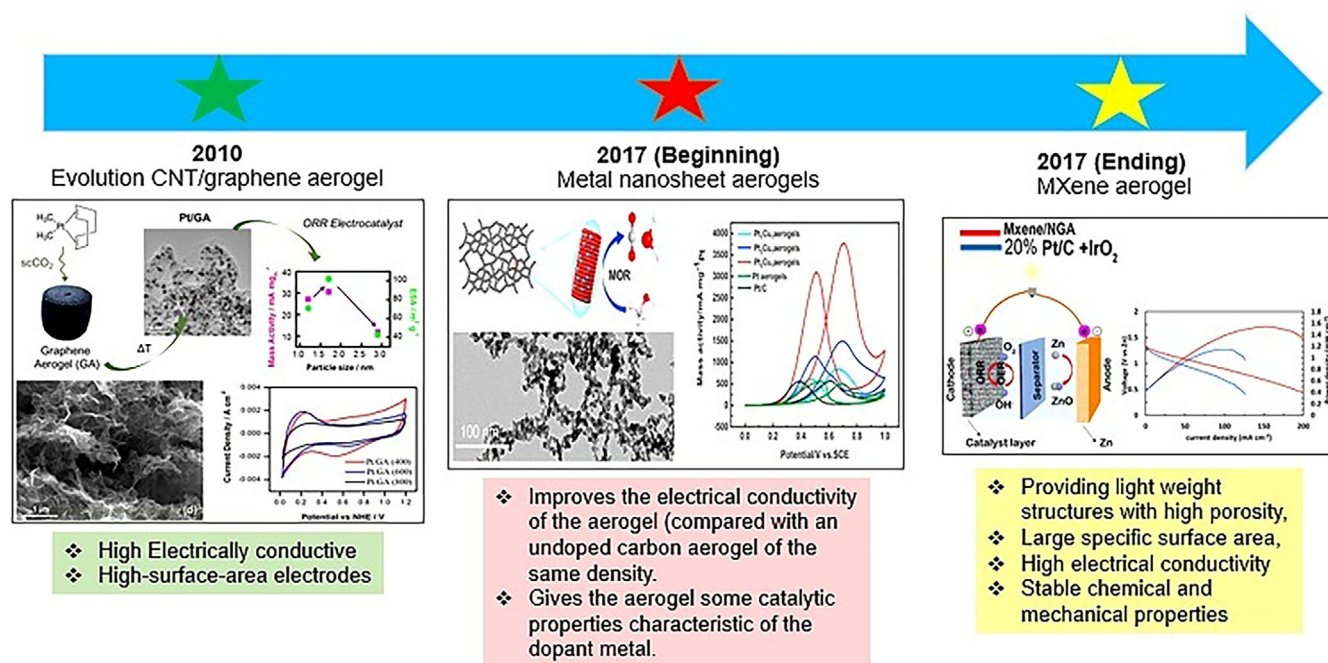


FIGURE 8 Milestones and benefits of organic aerogels in research fuel cell



FIGURE 9 Organic aerogel-based challenges

properties (molecular weight, composition, and degree of branching). In addition, structural parameters such as pore size, ordering, and even pore shape (regular or irregular) will strongly affect the final macroscopic performance of the aerogel materials.

Although the use of aerogel polymers is still less practiced in activity catalysts, based on the explanation above, this opportunity shines through the search for suitable raw materials with more economical sustainability factors, particularly for commodity applications. Particularly, polymer-based aerogels have not yet been emphasized in catalytic fuel cells, and there is a large opportunity for researchers to study them in-depth. Organic aerogels are generally known to be expensive, but with the support of aerogels, which are one of the ten most advanced structures on the market today with extraordinary properties, researchers are actively engaged in research.¹⁰⁵ Furthermore, the development of “green” technology was enabled using aerogel materials.

Carbon aerogels have demonstrated that their performance is much better than that of other materials in fuel cell applications.¹⁰⁶ Unfortunately, it faces limitations in the industry because its production output is still limited and not economic. In the case of aerogels, where research funding for laboratory-scale production is scarce, even the development of aerogels can be done at home, as recommended by the COVID-19 dynamic. This allows researchers to research from the comfort of their own homes with key safety features that must be prioritized. Simple preparation with autoclaving-related hydrothermal temperatures and freeze-drying process of aerogels can be developed successfully. Many media and internet sources support this by describing how to make aerogels at home. This increased attention shows not only that researchers noticed this but also public perception has become more aware of the existence of aerogels. Figure 9

recaps organic aerogel information that makes it simple to understand their behaviors.

Currently, the barrier factors of aerogel materials are:

- high manufacturing cost,
- inefficient mechanical properties,
- outcome of wellness

In this string barrier from the solvent exchange process and the low electronic conductivity, which is the parameter to the success of the commercialization of this material, time is also emphasized. As a solution, a synthetic strategy is being implemented in these challenges by reducing the solvent exchange time, which will be thoroughly investigated in the coming years. Furthermore, physical and chemical modification of the aerogel surface has the potential to improve the adsorption process in the future.

Once compared to other techniques, supercritical drying is the most luxurious and hazardous.¹⁰⁷ A highly desirable goal in aerogel preparation is the elimination of the supercritical drying process.¹⁰⁸ Dry eyes, dry skin, respiratory tract, and digestive system can all be affected by aerogel radiation. This is due to the silicosis respiration being influenced by the small particles. As a result, during the handling of aerogels, the use of protective equipment such as respirators, gloves, and goggles is highly recommended.¹⁰³

6 | CONCLUSIONS

Advances in organic aerogel fabrication are formulated in this review with specialization for the catalysis of low temperature fuel cells that yield high-performance

results. Organic aerogel development into catalytic low temperature fuel cells is focused on and concise in three main categories: carbon aerogels, carbon nanotube aerogels, and graphene aerogels. In addition, the high conductivity of organic aerogels provides benefit for catalytic fuel cell applications. Even though aerogel production is attentive to an application only on a laboratory scale, the nanostructured morphology and characterization can be carefully studied. These principal challenges in the commercialization of this material have great potential for industrial use. The development of organic aerogel development and deposition on aerogel surfaces in catalytic applications would receive a great deal of attention if the challenges presented in this paper can be addressed. This organic aerogel material will be a sustainable material in the future, particularly in catalytic low temperature fuel cell applications.

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DATA AVAILABILITY STATEMENT

Data openly available in a public repository that issues datasets with DOIs.

ORCID

Siti Kartom Kamarudin  <https://orcid.org/0000-0003-0383-5049>

Sahriah Basri  <https://orcid.org/0000-0002-2453-562X>

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